SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

20/20

ASSESSMENT TOOL 2 – DATA CLEANING CHALLENGE

Course Code:	**CSA0702**	**Course Title:**	**Computer Networks**
CO Assessed:	**CO1**	**Assessment:**	**Tool 2 – Concept Mapping**
Weightage:	***30%***	**Total Marks:**	**25 Marks**
CO1 Statement: ***CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)***			
Student Name:	P.Greeshma	**Reg. No.:**	192424418
Section / Batch:	AIDS - 2024	**Date of Submission:**	20/07/26
Faculty In-charge:	**Dr. R. S. Selvan**	**Project Mode:**	**Individual Submission**

Code of Conduct

I P.Greeshma (192424418) (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Anomaly Detection (5 Marks)	Cleaning Methods (5 Marks)	Data Transformation (5 Marks)	Analytical Comparison (5 Marks)	Governance Discussion (5 Marks)	Total (25 Marks)

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Le (Beg g)
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incor or missin relations ips
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable t apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

Comprehensive Analysis and Concept Mapping in Computer Vision & Image Processing

1. Complete the OSI Layer Concept Map

Concept Map

Application

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Presentation

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Session

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Transport

↓

Network

↓

Data Link

↓

Physical

Missing Layers

- Session Layer
- Data Link Layer

Analysis

Session Layer

- Establishes communication sessions between devices.
- Maintains synchronization.
- Controls dialogue between sender and receiver.
- Recovers interrupted sessions.

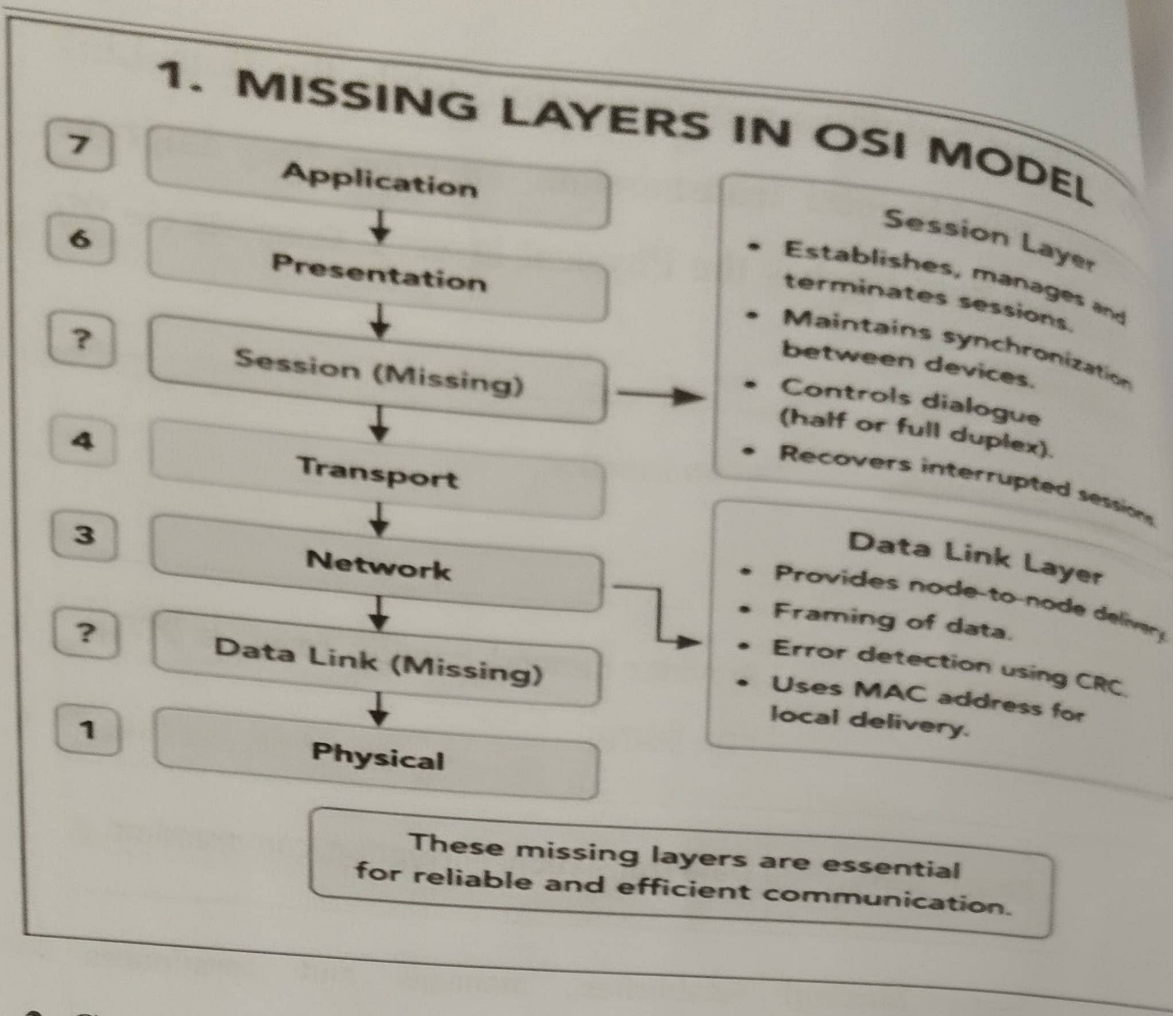
Data Link Layer

- Transfers data between directly connected devices.
- Performs framing.
- Detects errors using CRC.
- Uses MAC addresses for local delivery.

Conclusion

The Session layer maintains communication, while the Data Link layer ensures reliable local transmission. Together they improve reliability before data reaches the Physical layer. 1. Complete the OSI Layer Concept Map

OSI Layer Order	Layer Name	Main Function
7	Application	Provides network services to users (HTTP, FTP, SMTP)
6	Presentation	Data translation, encryption, compression
5	Session *(Missing)*	Establishes, manages and terminates sessions
4	Transport	End-to-end delivery, segmentation, error recovery
3	Network	Routing, logical addressing (IP)
2	Data Link *(Missing)*	Framing, MAC addressing, error detection
1	Physical	Transmits bits over cables or wireless media



2. Concept Map – Data Units and OSI Layers

Concept Map

Application

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Presentation

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Session

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Transport

│

└── **Segment**

↓

Network

│

└── **Packet**

↓

Data Link

│

└── **Frame**

↓

Physical

│

└── **Bits**

Analysis

During transmission, data undergoes encapsulation.

Application Data

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Segment (Transport Layer)

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Packet (Network Layer)

↓

Frame (Data Link Layer)

↓

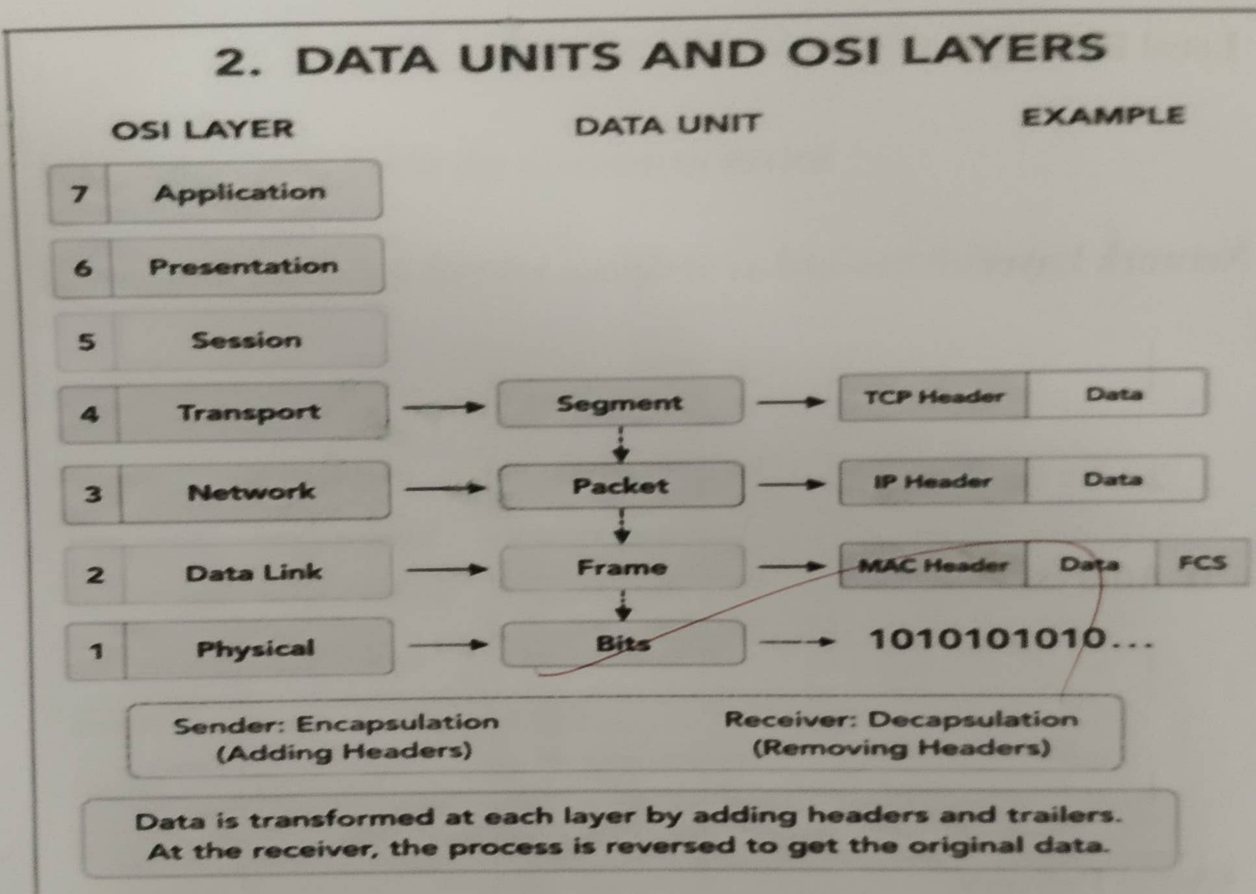
Bits (Physical Layer)

At the receiver, decapsulation converts Bits back into Frames, Packets, Segments, and finally Application Data.

This transformation enables efficient addressing, routing, error checking, and physical transmission.

OSI Layer	Data Unit	Function	Example

Application	Data	User information	Web page
Presentation	Data	Encryption/Compression	SSL/TLS
Session	Data	Session management	Login session
Transport	Segment	Segmentation, reliability	TCP Segment
Network	Packet	Routing using IP	IP Packet
Data Link	Frame	Framing, MAC addressing	Ethernet Frame
Physical	Bits	Binary transmission	10101010



3. MAC Address and IP Address Concept Map

Concept Map

Data Link Layer

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MAC Address

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↓

Local Delivery

Network Layer

|

↓

IP Address

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↓

Global Delivery

Analysis

- **MAC Address uniquely identifies a device inside a local network.**

- Switches use MAC addresses for communication.
- IP Address identifies devices across different networks.
- Routers use IP addresses to forward packets.

Together,

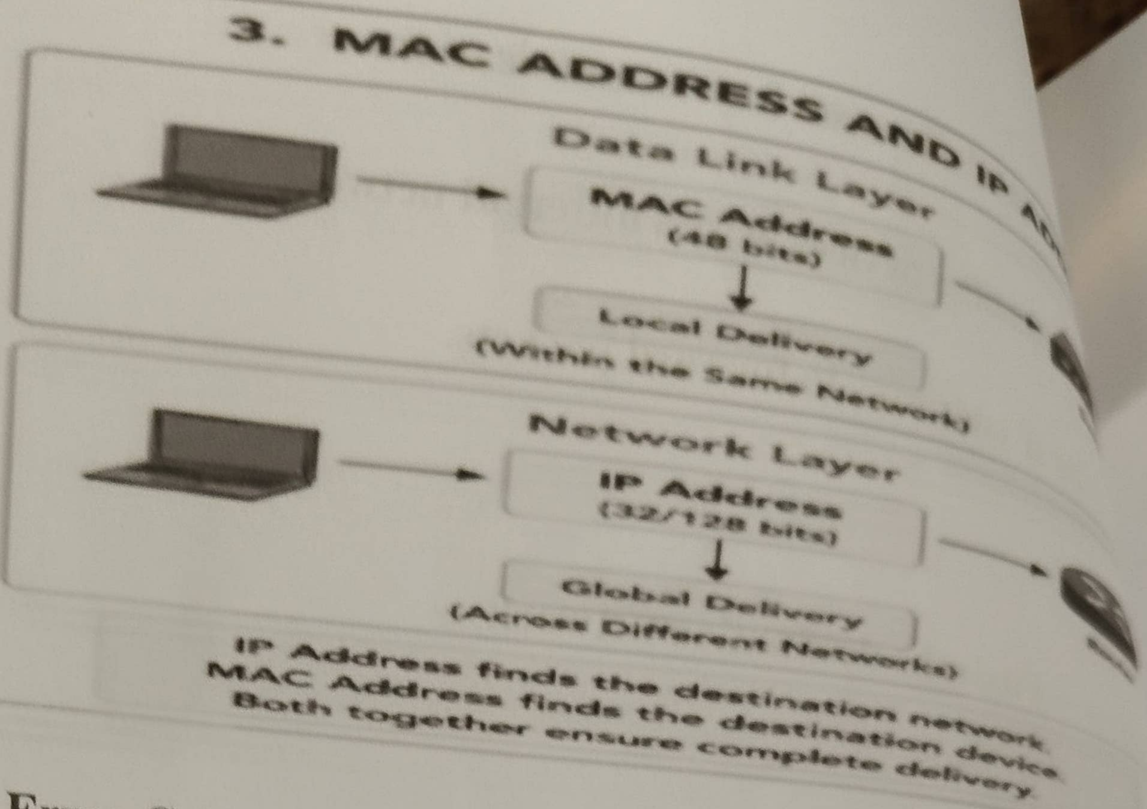
IP Address → Finds the destination network

↓

MAC Address → Finds the destination device

Hence both work together for complete end-to-end delivery.

Feature	MAC Address	IP Address
Layer	Data Link Layer	Network Layer
Address Type	Physical Address	Logical Address
Delivery	Local Network	Global Network
Used By	Switch	Router
Changes	Permanent	Can change
Example	00:1A:2B:3C:4D:5E	192.168.1.10



4. Error Control Concept Map

Concept Map

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        Error Control

      /        \

     /          \

Data Link Layer     Transport Layer

     |            |

     |            |

Error Detection    Error Correction

     |            |
```

Reliable Communication

Analysis

Data Link Layer

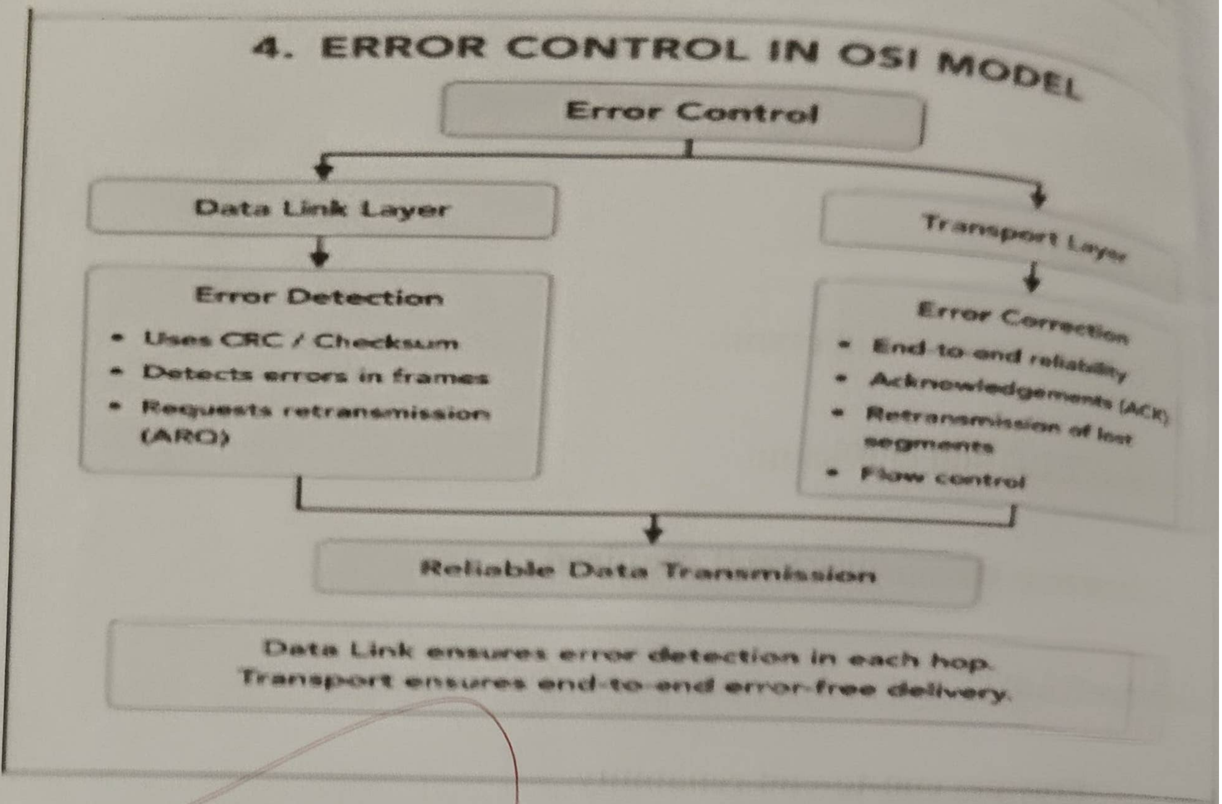
- **Detects transmission errors.**
- **Uses CRC and checksum.**
- **Requests retransmission if required.**

Transport Layer

- **Provides end-to-end reliability.**
- **Uses acknowledgements.**
- **Performs retransmissions.**
- **Controls flow.**

Together they reduce packet loss and ensure accurate data delivery.

Layer	Function	Techniques Used	Result
Data Link	Error Detection	CRC, Checksum	Detects frame errors
Transport	Error Correction	ACK, Retransmission, Flow Control	Reliable end-to-end



5. Concept Map – Web Browsing Using OSI Layers

OSI Layer	Function During Web Browsing	Protocol/Example
Application	Opens website	HTTP, HTTPS
Presentation	Encrypts/Decrypts data	SSL/TLS
Session	Creates communication session	Session Management
Transport	Reliable connection	TCP
Network	Routes packets	IP
Data Link	Frame transmission	Ethernet, Wi-Fi
Physical	Sends bits	Cable/Fiber/Radio

Concept Map

User Opens Website

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↓

Application

(HTTP/HTTPS)

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↓

Presentation

(Encryption/SSL)

|

↓

Session

(Session Management)

|

↓

(TCP)

|

↓

Network

(IP Routing)

|

↓

Data Link

(Frame & MAC)

|

↓

Physical

(Bits through Cable/Wi-Fi)

Analysis

Every layer performs a specific function.

If one layer fails:

- **Physical failure → No transmission.**
- **Data Link failure → Frames are lost.**
- **Network failure → No routing.**
- **Transport failure → TCP connection fails.**
- **Session failure → Communication ends.**
- **Presentation failure → Encryption/decryption fails.**
- **Application failure → Web page cannot load.**

Thus, all OSI layers depend on one another for successful web browsing